Higher-dimensional exponential metrics

A proposed axiomatic existence and uniqueness proof

Research manuscript

Abstract

We propose a proof of uniqueness, up to a deterministic multiplicative constant, for subcritical weak exponential metrics of the whole-space log-correlated Gaussian field in dimension $d \geq 3$. The construction and comparison results of Contreras Hip and Zhuang reduce the problem to identifying the optimal lower and upper comparison constants. Our main ingredient is a finite family of smooth annular field discounts which force a shortcut along every sufficiently substantial geodesic passage. We construct the discounts by attracting paths to a finite rectifiable graph whose marked arcs are unavoidable; dimension $d \geq 3$ is used to route this graph. A change-of-measure count then rules out a nontrivial comparison interval. The proposed uniqueness theorem yields a four-axiom characterization and convergence in probability for the convolution approximations covered by the construction theorem.

Status. This is an AI-assisted proposed research proof. The uniqueness argument and its consequences are not presented as established results. The accompanying audit record states the scope of the review and the changes from the manuscript dated 2 October 2026.

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1 Introduction

Fix an integer $d \geq 3$ and a parameter $\xi > 0$ for which the LFPP distance exponent satisfies $Q = Q(\xi) > \sqrt{2d}$. Let h be a centered whole-space log-correlated Gaussian field (LGF), viewed modulo additive constants. Thus, for smooth compactly supported test functions φ, ψ of integral zero,

$$\mathrm{Cov}((h,\varphi),(h,\psi)) = \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} \varphi(x)\psi(y)\log\frac{1}{|x-y|}\,dx\,dy\,.$$

This field is distinct from the Gaussian free field when $d \geq 3$. We write $h_r(z)$ for its spherical average and use the representative $h_1(0) = 0$ when a representative is needed. The natural distance normalization at location z and scale r is

$$s_r(z) = r^{\xi Q} e^{\xi h_r(z)}\,.$$

The metric assignments below take values in finite metrics which induce the Euclidean topology. For a metric D and an open set U, write $D(\cdot,\cdot;U)$ for the internal distance, obtained by taking the infimum of D-lengths of paths contained in U, with value $+\infty$ if there is no such rectifiable path. A metric assignment is a measurable map from $\mathcal{D}'(\mathbb{R}^d)$, with the sigma-algebra generated by test-function pairings, to continuous metrics, with the Borel structure of local uniform convergence. Write $e_1 = (1, 0, \ldots, 0)$.

Definition 1.1 (Strong metric)**.** An assignment $h \mapsto D_h$ satisfies the following axioms whenever h is an LGF plus a possibly dependent random continuous function.

(I) D_h is a length metric.

(II) For each deterministic open set U, the internal metric $D_h(\cdot,\cdot;U)$ is measurable with respect to $h|_U$.

(III) Almost surely, simultaneously for all continuous functions $f : \mathbb{R}^d \to \mathbb{R}$,

$$D_{h+f} = e^{\xi f}\cdot D_h\,.$$

Here the right side is obtained by minimizing the integral of $e^{\xi f}$ against D_h-arclength.

(IV) For each fixed $r > 0$ and $z \in \mathbb{R}^d$, almost surely for all $u, v \in \mathbb{R}^d$,

$$D_h(ru+z, rv+z) = D_{h(r\cdot+z)+Q\log r}(u,v)\,.$$

Theorem 1.2 (Proposed characterization)**.** *There exists an assignment satisfying Definition 1.1. Any two such assignments for the same (d,ξ,Q) differ almost surely by a deterministic positive multiplicative constant, simultaneously for all endpoint pairs and after every continuous perturbation of the field.*

For each convolution scheme in [1, Remark 1.1], the corresponding LFPP metrics, normalized by their point-to-point medians, converge in probability locally uniformly to this assignment with its median fixed to one. With the same deterministic normalizers, this convergence holds for each LGF plus a bounded continuous function.

The approximation statement concerns the convolution kernels obtained from smooth compactly supported radial white-noise kernels in [1, Remark 1.1]. It makes no assertion for other mollifiers. The parameter correspondence and independence of $Q(\xi)$ from the underlying white-noise kernel are as in [1, Section 1.2] and [6]. The axioms themselves do not require a choice of mollifier.

The planar subcritical characterization was proved by Gwynne and Miller [3]. The available higher-dimensional construction produces metrics with tightness across scales rather than exact dilation covariance. We first compare these weak metrics; exact covariance will follow from their proposed uniqueness.

Definition 1.3 (Weak metric)**.** A weak metric assignment satisfies Axioms (I)–(III), translation covariance, and the following scale condition. There are deterministic constants $\mathfrak{c}_r > 0$ such that, for each finite $R > 0$, the laws of

$$\mathfrak{c}_r^{-1} e^{-\xi h_r(0)} D_h(r\cdot, r\cdot)\,, \qquad 0 < r < R\,,$$

are tight for local uniform convergence, and every subsequential limit induces the Euclidean topology. If the continuous perturbation of the LGF is bounded, the same requirement holds for all $r > 0$.

Theorem 1.4 (Proposed weak uniqueness)**.** *Any two weak metric assignments with the same parameter ξ differ almost surely by a deterministic positive multiplicative constant, simultaneously for all endpoint pairs and after every continuous perturbation of the field.*

1.1 Construction and comparison inputs

Definition 1.3 is the weak metric class of [1, Section 1.2], with $\gamma = Q - \sqrt{Q^2 - 2d} \in (0, \sqrt{2d})$. We use the following results of that paper.

(E1) Every sequence of approximation scales tending to zero has a deterministic subsequence along which normalized LFPP converges in probability locally uniformly to a measurable weak assignment. One subsequence and assignment work for each LGF plus a bounded continuous function [1, Theorem 1.3].

(E2) The scale constants in Definition 1.3 may be chosen as $\mathfrak{c}_r = r^{\xi Q}$ [1, Theorem 1.6].

(E3) Two weak assignments satisfy a deterministic global bi-Lipschitz comparison [1, Proposition 4.6]. The local comparison criterion underlying that result is [1, Proposition 3.13]. Locality of the assignments gives joint locality; subtracting a spherical average and applying Weyl scaling gives joint ξ-additivity.

(E4) Events measurable modulo constants on geometrically separated concentric annuli satisfy the following bound: sufficiently high marginal probabilities give any prescribed exponential rate for the probability of more than a fixed positive fraction of failures [1, Lemma 2.7]. Appendix A gives a direct proof of the part used here.

(E5) If U is connected and open and $K \Subset U$ is compact and connected, normalized internal diameters have bounded p-moments, uniformly over scale, for

$$0 < p < \frac{2d}{\xi\gamma}\,.$$

This is [1, Proposition 4.11]. A singleton K is harmless, since its diameter is zero.

The construction in (E1) uses the tightness theorem of Ding, Gwynne and Zhuang [2]. None of these inputs asserts weak uniqueness.

1.2 Outline of the proof

Fix two weak assignments $D_h, \widetilde{D}_h$ and set

$$c_* = \sup\{c : \widetilde{D}_h \geq cD_h \text{ a.s.}\}\,, \qquad C_* = \inf\{C : \widetilde{D}_h \leq CD_h \text{ a.s.}\}\,.$$

All-pairs inequalities can be tested on a countable dense set. By continuity and (E3), the extremal bounds are attained and

$$0 < c_* \leq C_* < \infty\,, \qquad c_* D_h \leq \widetilde{D}_h \leq C_* D_h \quad \text{a.s.}$$

The same inequalities hold for path lengths, internal metrics, and common continuous Weyl reweightings. The task is to prove $c_* = C_*$.

Suppose that $c_* < C_*$. The definition of c_* supplies arcs along which the comparison ratio is close to its lower extreme. A frequency argument produces such arcs at many scales, and independent fine-field trials place them in finitely many prescribed boxes. The geometric problem is to make these favorable arcs useful to every sufficiently substantial geodesic passage through an annulus.

We solve this problem by joining the arcs into a finite rectifiable graph near a Euclidean sphere. Its marked arcs separate the graph, so a graph geodesic with sufficiently separated endpoints must traverse one of them. A deep, narrow field discount attracts minimizing paths to the graph. The needed conclusion is quantitative: a minimizing path can be connected to its limiting graph path at a cost negligible compared with the discounted graph length. This permits transfer of a strict comparison shortcut. Compactness and stability then reduce the configuration-dependent construction to a finite deterministic family of smooth discounts. The compactness, graph construction, and stability arguments occupy Sections 2–4; the scale and probability arguments occupy Sections 5 and 6.

The remaining step rules out a geodesic whose endpoint comparison ratio is close to C_*. Such a geodesic gives many favorable sets of discounts. After changing the field by the corresponding deterministic perturbation, the same sets are tested against one common base geodesic. The shortcut estimate and a bound on the time spent in each discount leave too few such sets. Sections 7 and 8 make this comparison using Cameron–Martin absolute continuity. The number of discounts is fixed before the scale ratio tends to zero. Section 9 combines the resulting estimate with the frequency of near-extreme ratios to obtain $c_* = C_*$. Section 10 then deduces exact covariance and the stated LFPP convergence.

2 Local compactness

Write $B_r(z)$ for the Euclidean open ball and $A_{a,b}(z) = B_b(z) \setminus \overline{B}_a(z)$ for the open annulus. Omitted centers are zero. We need compact families of local metric configurations so that a field discount constructed for one configuration continues to work for nearby configurations.

Lemma 2.1 (Continuous weights)**.** *Let (X, D) be a locally compact length space, and let $w : X \to (0, \infty)$ be continuous. Then $w \cdot D$ induces the same topology as D. A continuous path P on a compact interval is rectifiable for one of these metrics if and only if it is rectifiable for the other, and*

$$\operatorname{len}(P; w \cdot D) = \int_P w \, ds_D\,.$$

For an open set $U \subset X$,

$$(w \cdot D)(x, y; U) = \inf_{\substack{P: x \to y \\ P \subset U}} \int_P w \, ds_D \,,$$

where the infimum is over D-rectifiable paths. These conclusions hold locally for a metric coordinate whose sufficiently short distances are realized inside its domain of inspection. Common positive continuous weights preserve two-sided metric comparison.

Proof. For a D-rectifiable path, using each of its subpaths as a competitor gives the upper bound for weighted length. To prove the reverse bound, fix a neighborhood B on which w oscillates by at most δ and a smaller neighborhood $B' \Subset B$. The distance from $\overline{B}'$ to ∂B is positive. Consequently, for $x, y \in B'$ with $D(x, y)$ sufficiently small, a path leaving B pays at least $(\inf_B w) D(x, y)$ before its first exit, whereas a path remaining in B pays at least the same amount. Thus

$$(w \cdot D)(x, y) \geq (\inf_B w) D(x, y)$$

near the diagonal in $B' \times B'$. Short almost-minimizing D-paths give the corresponding local upper bound.

These comparisons prove nondegeneracy and equality of topologies. Cover the compact image of P by finitely many such neighborhoods and subdivide its parameter interval accordingly. On each piece, refine partitions until the local comparison applies. Summing gives

$$\operatorname{len}(P; w \cdot D) \geq \int_P w \, ds_D - \delta \operatorname{len}(P; D) \,.$$

Let $\delta \downarrow 0$. The same partition argument, using the minimum of finitely many positive lower bounds for w, gives

$$\operatorname{len}(P; w \cdot D) \geq m_P \operatorname{len}(P; D)$$

for some $m_P > 0$, allowing infinite lengths. This proves the converse rectifiability assertion. Taking infima over paths contained in U proves the internal-metric identity.

The argument is local and therefore also applies to the stated metric coordinates. Finally, if $mD \leq E \leq MD$, the same inequalities hold for the arclength measures of every rectifiable path. Integrating a common weight and taking infima yields

$$m(w \cdot D) \leq w \cdot E \leq M(w \cdot D) \,.$$

□

2.1 Metric configurations

Lemma 2.2 (Limits of internal metrics)**.** *Suppose continuous length metrics D_n on $\mathbb{R}^d$ converge locally uniformly to a metric D inducing the Euclidean topology. Then D is locally a length metric. For every connected open U, the metrics $D_n(\cdot, \cdot; U)$ are relatively compact for local uniform convergence on $U \times U$. Each limit is finite, induces the Euclidean topology, is locally a length metric, and gives the same lengths as D to paths in U.*

The same conclusions hold for jointly convergent internal coordinates on nested domains when common positive lower moduli and local upper moduli are retained. In particular, if $\overline{U}_1 \subset U$ is compact, $x_n \to x$, $y_n \to y$, and $D_n(x_n, y_n; U_1) \leq A$, a joint limit contains a path from x to y in $\overline{U}_1$ of length at most A for the U coordinate.

Proof. Fix compact balls $K \subset \operatorname{int} K'$ inside the inspected domain. The D-distance from K to $\partial K'$ has a positive minimum, and local uniform convergence gives a common positive bound for D_n. Sufficiently short D_n-almost-minimizers starting in K stay in K'. Parameterize them by length and extend constantly to a fixed time interval. The common lower modulus gives Euclidean equicontinuity. Arzelà–Ascoli and lower semicontinuity of length then give short limiting paths in K'. This proves the local length property.

A connected open subset of a locally length space is rectifiably connected: the set of points reachable by finite-length paths is both open and relatively closed. Fix a finite-length D-path inside U and subdivide it into pieces confined to balls compactly contained in U. Replacing these pieces by short D_n-almost-minimizers bounds the internal distances of their endpoints. Finitely many such paths and local neighborhoods give uniform bounds and a common upper modulus on each compact subset of $U \times U$. Arzelà–Ascoli on a compact exhaustion gives relative compactness.

Every internal coordinate dominates the corresponding global metric, so its limit retains a positive lower modulus. Near each diagonal point, short global competitors cannot leave U. Hence the global and internal distances agree locally, as do their limits. Equality of the lengths of paths in U follows by subdivision.

For the final assertion, take U_1-competitors of length at most $A + 1/n$. Their images lie in the fixed compact set $\overline{U}_1 \subset U$. Reparameterization and the common lower modulus give a uniformly convergent subsequence. Its limit has the asserted endpoints and length by lower semicontinuity. It may meet ∂U_1; it remains inside U. □

We use the nested domains

$$U = A_{1/2,5}\,, \qquad U_1 = A_{3/4,7/2}\,, \qquad \overline{U}_1 \subset U\,.$$

A configuration consists of the two normalized internal metrics on each of U, U_1. Inputs (E2) and Lemma 2.2 give joint tightness, uniformly over scale. By imposing common upper bounds, upper moduli, and positive lower moduli on a countable compact exhaustion, we obtain compact sets $\mathcal{C}$ of arbitrarily high probability. We take the closure in the product local uniform topology. The closure retains local length, comparison, and agreement of path lengths on overlapping domains. All neighborhoods used below are relative to such a compact set $\mathcal{C}$.

Lemma 2.3 (Local realization)**.** *Fix $K \Subset O \Subset U$, with K compact and O open. There is $\delta_{\mathrm{loc}} > 0$ such that, for every configuration in $\mathcal{C}$ and either U coordinate D^U, any $x, y \in K$ with $D^U(x,y) < \delta_{\mathrm{loc}}$ are joined by a path in O of D^U-length exactly $D^U(x,y)$. The analogous assertion holds for U_1.*

Proof. The common lower modulus gives $a > 0$ such that the distance from K to ∂O is at least a in every original coordinate. Set $\delta_{\mathrm{loc}} = a/4$. Approximate a coordinate in the closure by original internal metrics D_n^U. For $D^U(x,y) < a/4$, almost-minimizers of $D_n^U(x,y)$ have length less than $a/2$ for large n, so they stay in O. Compactness in $\overline{O}$, followed by lower semicontinuity, gives a limiting path realizing $D^U(x,y)$. It cannot meet ∂O, since any such path would have length at least a. The choices are uniform in $\mathcal{C}$. □

Lemma 2.4 (Path recovery)**.** *Fix $K \Subset O \Subset U$, a positive continuous weight w, and corresponding metric coordinates D, D' of two configurations in $\mathcal{C}$. Set*

$$\epsilon = \sup_{x,y \in \overline{O}} |D'(x,y) - D(x,y)|\,, \qquad M = \max_{\overline{O}} w\,.$$

There is a common nondecreasing inverse modulus χ, tending to zero at zero, such that $D''(x,y) \le s$ implies $|x-y| \le \chi(s)$ on $\overline{O}$ for all coordinates D'' in $\mathcal{C}$. Put

$$\omega_w(t) = \sup_{\substack{x,y\in\overline{O}\\ |x-y|\le t}} |w(x)-w(y)|\,.$$

If $\epsilon \le \delta$ and $2\delta < \delta_{\mathrm{loc}}$, every D-rectifiable path $P \subset K$ of length at most L has a replacement $P' \subset O$ with the same endpoints and

$$\operatorname{len}(P';D') \le L + (L/\delta+1)\epsilon\,, \tag{1}$$

$$\int_{P'} w\,ds_{D'} \le \int_P w\,ds_D + 2L\omega_w(\chi(2\delta)) + 2M(L/\delta+1)\epsilon\,. \tag{2}$$

The paths can be parameterized on the same interval with pointwise Euclidean distance at most $2\chi(2\delta)$.

Proof. The inverse modulus follows from compactness: otherwise a sequence of pairs with metric distance tending to zero and Euclidean separation bounded below would converge to distinct points at zero distance for a limiting coordinate.

Subdivide P by D-arclength into at most $L/\delta+1$ pieces of lengths $\ell_i \le \delta$, with endpoints z_i, z_{i+1}. Lemma 2.3 gives D'-geodesics P_i' in O of length at most $\ell_i+\epsilon \le 2\delta$. Their concatenation proves (1). Every point of either the old or the new piece lies within $\chi(2\delta)$ of z_i. Writing $\Omega = \omega_w(\chi(2\delta))$, we have

$$\int_{P_i'} w\,ds_{D'} \le (w(z_i)+\Omega)(\ell_i+\epsilon)\,, \qquad \int_{P|_i} w\,ds_D \ge (w(z_i)-\Omega)\ell_i\,.$$

Subtract, sum, and use $w(z_i)+\Omega \le 2M$ to obtain (2). Parameterizing corresponding pieces on the same subinterval gives the pointwise bound. Constant pieces are retained. □

If $D_n \to D$ locally uniformly, choosing $\delta_n = \sqrt{\epsilon_n}+1/n$ in Lemma 2.4 gives uniformly convergent replacements with weighted upper limit no greater than the original weighted length. The error is uniform over paths of bounded length in a fixed compact set. Moving endpoints can be joined by short locally realized paths, whose costs vanish by the common upper modulus.

Dubédat and Falconet [8, Lemma 7.1] prove a related fixed-weight continuity statement for intrinsic metrics on $[0,1]^2$ with common two-sided Euclidean moduli. Their conclusion is pointwise convergence of the reweighted distances. We need the following local form, including uniformity over the prescribed endpoint pairs and convergence of minimizing paths under compact confinement.

Lemma 2.5 (Fixed-weight continuity)**.** *Suppose configurations converge in $\mathcal{C}$, and fix a positive continuous weight w. If the endpoints, all relevant minimizing paths, and recovering competitors are confined to a common compact subset of the inspected domain, their weighted distances converge uniformly over the endpoint pairs under consideration. Limits of constant-speed weighted minimizing paths with bounded lengths are weighted minimizing paths for the limiting local structure.*

Proof. On the confining compact set, the weight has a positive minimum. Weighted length bounds therefore bound original lengths, giving compactness of paths. Subdivision into pieces on which w has small oscillation gives weighted lower semicontinuity from ordinary lower semicontinuity of length. Lemma 2.4 gives the upper bound by recovering a limiting competitor in a slightly larger compact neighborhood. Short connections handle moving endpoints. A subsequence argument gives uniform convergence of distances and identifies limits of minimizers. □

2.2 A uniform upper modulus

The final comparison will discretize arbitrary endpoints. We need its modulus estimate uniformly over the macroscopic scale. We derive this directly from the diameter moments, without assuming independent spherical-average increments.

Lemma 2.6 (Upper modulus)**.** *For each $0 < \alpha < \xi(Q - \sqrt{2d})$ and finite $L > 0$, there is $a > 0$ such that*

$$\mathbb{P}\left[\sup_{\substack{x,y\in B_{LR}\\ |x-y|\le\varepsilon R}} D_h(x,y) > \varepsilon^{\alpha} s_R(0)\right] = O(\varepsilon^{a})$$

uniformly in $R > 0$. The same argument bounds internal diameters of small balls in a fixed enlarged macroscopic ball.

Proof. Set $t = \varepsilon R$, fix $z \in B_{(L+2)R}$, and write

$$X = h_{2t}(z) - h_R(0)\,, \qquad H = h - h_{2t}(z)\,.$$

Let σ, τ be uniform probability measures on $\partial B_{2t}(z), \partial B_R$, respectively, and put $F_\mu(w) = \int \log|w-u|^{-1}\mu(du)$. Then

$$k(w) := \operatorname{Cov}(X, H(w)) = F_\sigma(w) - F_\tau(w) - \int (F_\sigma - F_\tau)\, d\sigma\,.$$

This denotes covariance with the restricted generalized field. The logarithmic potential of a sphere is continuous, including on the sphere: the integral over a cap of radius δ is $O(\delta^{d-1}(1 + |\log\delta|))$, uniformly in the cap center. Scaling the two sphere potentials therefore gives

$$\operatorname{Var} X = \log(R/t) + O_L(1)\,, \qquad \sup_{B_{2t}(z)} |k| = O_L(1)\,.$$

In the second estimate, the $\log(1/t)$ terms of the small-sphere potentials cancel.

Define

$$M(H) = (2t)^{-\xi Q} \sup_{u,v\in\overline{B}_t(z)} D_H(u, v; B_{2t}(z))\,.$$

Locality makes this a functional of the restricted field. Input (E5) and translation covariance give $\mathbb{E}[M(H)^p] \le C_p$, uniformly in z, t, for $0 < p < 2d/(\xi\gamma)$. Gaussian tilting and local Weyl comparison yield

$$\begin{aligned}\mathbb{E}[e^{\xi p X} M(H)^p] &= e^{\xi^2 p^2 \operatorname{Var}(X)/2}\mathbb{E}[M(H + \xi p k)^p]\\ &\le C_{p,L} e^{\xi^2 p^2 \operatorname{Var}(X)/2}\,.\end{aligned}$$

Apply the tilt first to bounded truncations and then use monotone convergence. Only the bound for k on $B_{2t}(z)$ enters the comparison. It follows that

$$\mathbb{E}\left[\left(s_R(0)^{-1} \sup_{u,v\in\overline{B}_t(z)} D_h(u, v; B_{2t}(z))\right)^p\right] \le C_{p,L}\varepsilon^{\xi Q p - \xi^2 p^2/2}\,.$$

For $s \in (0, \xi Q)$, take $p = (\xi Q - s)/\xi^2$. This moment is admissible because $Q/\xi < 2d/(\xi\gamma)$, equivalently $\gamma^2 < 2d$. Markov's inequality gives exponent $(\xi Q - s)^2/(2\xi^2)$ for the probability that the diameter exceeds $\varepsilon^s s_R(0)$.

Choose $\alpha < s < \xi(Q - \sqrt{2d})$, so that this exponent exceeds d. At scale $2^{-j}\varepsilon R$, use a Euclidean grid of spacing $2^{-j}\varepsilon R/(4d)$ in a fixed enlarged macroscopic ball. A union bound over $O((2^{-j}\varepsilon)^{-d})$ centers, and then over $j \ge 0$, has positive polynomial decay. Each pair under consideration lies in a grid ball at a comparable scale. The difference $s - \alpha$ absorbs fixed constants. □

2.3 Geodesics

Lemma 2.7 (Properness and confinement)**.** *For a whole-space LGF, D_h and $\widetilde{D}_h$ are almost surely proper. For each finite $L > 0$ and $\delta > 0$, there is $A < \infty$ such that, uniformly in $R > 0$, with probability at least $1 - \delta$, every geodesic for either metric with endpoints in B_{LR} stays in B_{AR}.*

Proof. Normalized diameters of B_{LR} are tight. Write the crossing distance of $A_{tR/2,tR}$, divided by $s_R(0)$, as

$$(t/2)^{\xi Q} e^{\xi(h_{tR/2}(0) - h_R(0))} Z_{tR/2}\,.$$

The variables Z_r are bounded away from zero in probability, uniformly in r, by scale tightness and nondegeneracy. The Gaussian increment has variance at most $\log t + O(1)$. Since $Q > 0$, the displayed quantity tends to infinity in probability, uniformly in R. This uses no independence of its factors. A path leaving a sufficiently large ball must pay this crossing cost, which proves confinement for minimizing paths.

For properness, take $R = 1$ and a deterministic sequence $t_k \uparrow \infty$ so fast that the probabilities of crossing cost less than k are summable. Borel–Cantelli shows that distances from any fixed Euclidean compact set to ∂B_{t_k} tend to infinity. Closed metric balls are therefore Euclidean bounded and closed, hence compact. A proper length space is geodesic. The comparison of D_h and $\widetilde{D}_h$ gives the assertions for both metrics. □

We fix a measurable choice P_h of a constant-speed D_h-geodesic between each pair of endpoints, unchanged by multiplication of the metric by a scalar. One construction is to consider, for each positive integer n, the paths in $\overline{B}_n$ joining the endpoints and satisfying

$$D(P(s), P(t)) = |s - t| D(x, y)\,, \qquad s, t \in [0, 1] \cap \mathbb{Q}\,.$$

These fibers are compact, their graph is Borel, and some fiber is nonempty by properness. Choose the least such n and then a fixed measurable selector of its compact path set. The fibers and the least n are invariant under scalar multiplication.

All events below are taken in completed sigma-algebras. Path witnesses can be expressed using a compact exhaustion and bounds on their metric Lipschitz constants. Their projections are universally measurable; strict inequalities can be handled by countable unions. Excursion intervals are enumerated by rational times, and first and last visits to a closed set are measurable. Assertions about every geodesic will follow from deterministic statements, rather than from a choice of a geodesic selector.

3 Attraction to a finite graph

A finite rectifiable graph in a metric space (X, D) is a compact embedded finite topological graph whose edges are simple D-rectifiable arcs. When K is connected, give each edge its D-length and denote the resulting finite intrinsic metric by d_K. Then (K, d_K) is compact, d_K induces the inherited topology, and ambient and intrinsic lengths agree for paths in K.

We will make travel near such a graph much cheaper than travel elsewhere. The next lemma controls both the limiting path and the cost of joining the approximating path to its limit. The latter control is needed to transfer a shortcut from the graph back to the metric.

Lemma 3.1 (Collapse onto a graph)**.** *Let (X, D) be a proper length space and let $K \subset X$ be a connected finite rectifiable graph. Suppose $\lambda_n \downarrow 0$, $\eta_n/\lambda_n \to 0$, and continuous weights satisfy*

$$\lambda_n \le w_n \le 1\,, \qquad w_n|_K = \lambda_n\,, \qquad w_n \ge a_0 > 0 \text{ outside } N^D_{\eta_n}(K)\,.$$

Set $D_n = w_n \cdot D$. Let x_n, y_n lie in the closed η_n-tube of K, and choose $u_n, v_n \in K$ at D-distance at most η_n from them. If $u_n \to u$ and $v_n \to v$, then

$$\lambda_n^{-1} D_n(x_n, y_n) \longrightarrow d_K(u, v)\,.$$

Parameterize a D_n-geodesic P_n from x_n to y_n at speed λ_n and extend it constantly to a common interval. Every such sequence has a subsequence converging uniformly in D to a unit-speed d_K-geodesic P from u to v, with the same constant extension. Along this subsequence,

$$\sup_t \lambda_n^{-1} D_n(P_n(t), P(t)) \longrightarrow 0\,. \tag{3}$$

The same assertions hold when D is a metric coordinate on an open subset of $\mathbb{R}^d$ with the local realization property of Lemma 2.3, provided the minimizing paths, almost-minimizing competitors, access paths, and graph competitors are confined to a fixed compact subset of that domain.

Proof. Access paths and a shortest graph path give

$$L_n := \lambda_n^{-1} D_n(x_n, y_n) \le d_K(u_n, v_n) + 2\eta_n/\lambda_n\,. \tag{4}$$

Since $w_n \ge \lambda_n$, the path P_n is 1-Lipschitz for D. Its D-length outside the tube is at most $\lambda_n L_n/a_0$, so its maximum distance from K tends to zero. Properness, or the stated local confinement, gives a uniformly convergent subsequence with limit $P \subset K$. Lower semicontinuity and (4) give

$$d_K(u, v) \le \operatorname{len}(P; D) \le \liminf_n \operatorname{len}(P_n; D) \le \liminf_n L_n \le \limsup_n L_n \le d_K(u, v)\,.$$

Every inequality is an equality. The 1-Lipschitz limit realizes the graph distance in exactly that time.

Let T_n be the amount of normalized weighted time spent outside the tube before the constant extension. By Lemma 2.1, weighted arclength is $\lambda_n\, dt$. Original arclength is at most dt everywhere and at most $(\lambda_n/a_0)\, dt$ outside the tube. Hence

$$\operatorname{len}(P_n; D) \le L_n - (1 - \lambda_n/a_0) T_n\,.$$

The equality of limits just proved implies $T_n \to 0$. From a point $P_n(t)$ outside the tube, follow its excursion to the tube, at weighted cost at most $\lambda_n T_n$, and then an access path to some $q_n(t) \in K$, at cost at most $\eta_n + o(\eta_n)$. Uniform convergence gives

$$\sup_t D(q_n(t), P(t)) \to 0\,.$$

The identity from (K, d_K) to (K, D) has uniformly continuous inverse. Thus the same convergence holds with d_K in place of D. Travel from $q_n(t)$ to $P(t)$ along the graph, where the weight is λ_n, and divide the resulting bound by λ_n to obtain (3). □

Lemma 3.2 (Transferring a shortcut)**.** *Retain the setting and notation of Lemma 3.1. Let E be a metric on X and let $0 < m \le M < \infty$ satisfy $mD \le E \le MD$. Fix $b > 0$. Let $\sigma_i \subset K$ be finitely many marked arcs with endpoints u_i, v_i such that*

$$D(u_i, v_i) \ge a_1 > 0\,, \qquad \operatorname{len}(\sigma_i; E) \le c_0 D(u_i, v_i)\,, \qquad 0 < c_0 < c_1 < M\,.$$

Assume every graph geodesic with endpoint D-distance at least b traverses a complete marked arc. For all sufficiently large n, every D_n-geodesic between points in the closed η_n-tube with endpoint D-distance at least b contains a subsegment of D_n-length at least $\lambda_n a_1/2$ whose endpoints are joined by a path of $w_n \cdot E$-length at most c_1 times that length. In the local version, this shortcut stays in the confining domain.

Proof. Otherwise take a sequence of failing geodesics. Lemma 3.1 gives a limiting graph geodesic which traverses some σ_i. At its entrance and exit times, (3) gives concrete paths of D_n-cost $o(\lambda_n)$ from the approximating points to u_i, v_i. Their weighted E-costs are also $o(\lambda_n)$ by comparison. Joining these paths through σ_i gives a shortcut of cost at most

$$\lambda_n \operatorname{len}(\sigma_i; E) + o(\lambda_n) .$$

The intervening geodesic length, divided by λ_n, tends to the graph length of this arc, which is at least $D(u_i, v_i)$. The strict gap $c_0 < c_1$ contradicts failure. All connection paths come from the local construction in Lemma 3.1, so they stay in the confining domain when required. □

4 Routing marked arcs

To use Lemma 3.2, we need a graph on which every path joining sufficiently separated points crosses a marked arc. Dimension $d \geq 3$ enters here. A relatively closed subset of covering dimension at most $d-2$ cannot disconnect a connected Euclidean open set; see the classical nonseparation theorem recalled in [5, p. 1]. A finite closed union of simple arcs has covering dimension at most one. This statement applies to wild arcs and does not require a bound on their Euclidean Hausdorff dimensions.

Throughout this section, (D, E) is the pair of U coordinates of a configuration in $\mathcal{C}$, where $U = A_{1/2,5}$. In particular, these metrics induce the Euclidean topology on U, satisfy $mD \leq E \leq MD$ with $m = c_*$ and $M = C_*$, and have the local realization property of Lemma 2.3. Path lengths refer to these coordinates. The corresponding U_1 coordinates will supply access paths in the application of Section 6.

Lemma 4.1 (An embedded graph)**.** *Fix a sufficiently fine finite simplicial mesh of ∂B_2 and a spanning tree of its edge graph. Choose disjoint small vertex neighborhoods and thin edge tubes, disjoint away from incident vertex neighborhoods. In a central compact box of each tree edge e, disjoint from all other tubes, suppose there is a simple rectifiable arc γ_e such that*

$$D(\gamma_e(0), \gamma_e(1)) \geq a_1 , \quad \operatorname{len}(\gamma_e; D) \leq A_1 , \quad \operatorname{len}(\gamma_e; E) \leq c_2 D(\gamma_e(0), \gamma_e(1)) ,$$

where $0 < c_2 < c_3$ and $a_1, A_1, b > 0$. Suppose these regions lie in a thin open neighborhood V of ∂B_2.

The vertices and arcs can be joined into a connected finite rectifiable graph $K \subset V$ with a marked middle subarc σ_e of each γ_e, whose endpoint D-distance is at least $a_1/2$ and whose E-length is at most c_3 times that distance. Removing the relative interiors of the marked arcs leaves components belonging to single mesh vertices and their incident stubs. If each enlarged vertex star, including its incident edge tubes, has D-diameter less than $b/8$, every graph geodesic with endpoint D-distance at least $b/4$ traverses a complete marked arc.

Proof. Parameterize γ_e by D-arclength on $[0, L_e]$. Choose $\tau > 0$ with

$$2\tau < (1 - c_2/c_3)a_1 , \qquad 2\tau < a_1/2 ,$$

and mark $\sigma_e = \gamma_e([\tau, L_e - \tau])$ before routing any connector. If $a_e = D(\gamma_e(0), \gamma_e(L_e))$, then

$$D(\sigma_e(0), \sigma_e(1)) \geq a_e - 2\tau > a_1/2 , \qquad \operatorname{len}(\sigma_e; E) \leq c_2 a_e < c_3(a_e - 2\tau) .$$

Attachments are permitted only on $\gamma_e([0, \tau))$ and $\gamma_e((L_e - \tau, L_e])$.

Process the tree edges successively. At each stage, the union is a finite embedded rectifiable graph. After deleting marked interiors, each component already attached to a vertex contains exactly that vertex and stays in its enlarged star. Unattached stubs remain separate. Choose the left half-tube, an open enlargement of the central compact box inside its tube, and the left vertex neighborhood as overlapping connected open pieces. Their union is the left routing region. It excludes the opposite vertex neighborhood and other central boxes. Previously processed tubes meet it only in the designated vertex neighborhood. Choose the right routing region in the same way.

Remove $\gamma_e([\tau, L_e])$ from the left routing region. This relatively closed set has covering dimension at most one, so the remaining open set is connected by nonseparation. The local length property makes it rectifiably connected. Join the left vertex to $\gamma_e(0)$ by a finite-length path. Take its first contact with γ_e and its last preceding contact with the existing left attachment component. The intervening segment meets neither attachment set in its interior and ends in $\gamma_e([0, \tau))$.

Replace this segment by a shortest path in its compact image. Such a minimizer exists by constant-speed reparameterization, compactness, and lower semicontinuity; it is simple because its endpoints are distinct. It still avoids both attachment sets in its interior. All other existing pieces are excluded by the routing geometry. The connector therefore attaches to the old graph at only its two endpoints.

For the right connector, remove both the first connector and $\gamma_e([0, L_e - \tau])$ from the right routing region. Their union is relatively closed and has covering dimension at most one. It contains neither the right vertex nor $\gamma_e(L_e)$. Repeat the same construction, trimming at the last contact with the right attachment component and the first contact with γ_e. This preserves the invariant.

At the end, every stub is attached and the spanning tree makes K connected. A simple path which traverses no marked arc in full either remains within one marked arc or meets only one vertex attachment component. In the latter case its initial and terminal portions lie in incident partial marked arcs. Its endpoints therefore belong to one enlarged vertex star and have distance less than $b/8$. A graph geodesic is simple, so separation at least $b/4$ forces a complete marked-arc traversal. □

Confinement by collars. We record the estimate used to pass from this graph to a local metric perturbation. Let $\overline{V} \subset A \Subset U$ and suppose

$$D^U(\overline{V}, \partial A) \geq q > 0\,, \qquad w = 1 \text{ outside } V\,.$$

A path starting in $\overline{V}$ which reaches ∂A pays at least q after its last visit to $\overline{V}$ before its first exit. Consequently a collector-to-collector path of weighted length at most $\ell < q$ cannot leave A. For any point z on this path outside $\overline{V}$, its portion since the last collector visit has original length at most ℓ, so

$$D^U(z, \partial A) \geq q - \ell\,.$$

The same lower bound is immediate when $z \in \overline{V}$. A common upper modulus on $\overline{A}$ then gives a uniform positive Euclidean distance from these paths to ∂A. This proves compact confinement, including along convergent metric configurations. For collector endpoints, restricting weighted path competitors from U to A does not change a strict distance test below q. At an actual global metric realization, the same first-exit argument applies to paths in $\mathbb{R}^d$. Thus a threshold $t < q$ gives equivalent global and internal tests, both for $< t$ and for its negation $\geq t$.

Lemma 4.2 (A smooth annular discount)**.** *Suppose the graph of Lemma 4.1 lies in $V \Subset A_{1,3}$, and every point of $\overline{V}$ has an access path to a mesh vertex of D-length at most ϵ_0. Suppose positive*

collar bounds confine sufficiently short paths between points of $\overline{V}$ to $A_{1,3}$. Fix $c_3 < c' < M$. If ϵ_0 is sufficiently small compared with b and the collar bounds, there is a nonnegative $f \in C_c^\infty(A_{1,3})$, positive exactly on V, such that:

(i) any two points of $\overline{V}$ can be joined at $e^{-\xi f} \cdot D$-cost less than $3\epsilon_0$;

(ii) every minimizing path between points of $\overline{V}$ whose original endpoint D-distance is at least b contains a subsegment of weighted length at least $c > 0$ whose endpoints are joined inside $A_{1,3}$ by a path of $e^{-\xi f} \cdot E$-length at most c' times that length.

One can first obtain (ii) *with endpoint threshold $b/2$, length $2c$, and ratio strictly smaller than c'. With these margins, the final assertions hold throughout a relative neighborhood of the configuration in $\mathcal{C}$.*

Proof. Step 1: Construction of the weight. Choose smooth $0 \le f_0 \le 1$ positive exactly on V. Let $\lambda_n \downarrow 0$, $\eta_n = o(\lambda_n)$, and choose smooth $0 \le \varphi_n \le 1$ equal to one near K, with support in $N_{\eta_n}^D(K) \Subset V$. This is possible because D induces the Euclidean topology. For $M_n = \xi^{-1} \log \lambda_n^{-1} > 1$, set

$$f_n = f_0 + (M_n - f_0)\varphi_n\,, \qquad w_n = e^{-\xi f_n}\,.$$

These weights satisfy Lemma 3.1 with $a_0 = e^{-\xi}$. Access paths and graph paths cost at most

$$2\epsilon_0 + \lambda_n \operatorname{diam}(K, d_K)\,.$$

Take $\epsilon_0 < e^{-\xi} b/200$ small enough for the collars. For all sufficiently large n,

$$\lambda_n \operatorname{diam}(K, d_K) \le \epsilon_0/2\,.$$

The competitors then cost at most $5\epsilon_0/2$, leaving a uniform strict margin below $3\epsilon_0$. This proves (i) and compactly confines all minimizing paths under consideration. Increasing n further preserves this margin.

Step 2: Transfer of the graph shortcut. A minimizer with original endpoint distance at least $b/2$ must hit the tube; otherwise its weighted cost is at least $a_0 b/2 > 3\epsilon_0$. Its original lengths before the first tube hit and after the last tube hit sum to at most $3a_0^{-1}\epsilon_0 < b/8$. Its first and last tube points therefore have original distance at least $3b/8$. Apply Lemma 3.2 to the intervening minimizing segment, with threshold $b/4$ and a ratio between c_3 and c'. Fix one sufficiently large n and then reduce $c > 0$ to reserve the stated margins.

Step 3: Stability for a fixed discount. Fix $f = f_n$, put $w = e^{-\xi f}$, and suppose $(D_k, E_k) \to (D, E)$ in $\mathcal{C}$. The access-and-graph competitors have original lengths bounded by

$$L_0 = 2\epsilon_0 + \operatorname{diam}(K, d_K)\,.$$

They lie in a common compact subset of $A_{1,3}$. Indeed, each access path has length at most ϵ_0, and the collar estimate above gives a uniform boundary buffer; the graph is already compactly contained in V. Lemma 2.4 recovers these competitors for D_k with an error tending to zero uniformly over all collector endpoints. Their weighted costs are therefore less than $4\epsilon_0$ for large k.

The collar bounds persist with a positive margin. Applying the quantitative confinement estimate above to D_k shows that every sufficiently short weighted minimizer stays in a common compact subannulus. Lemma 2.5 now applies, without having assumed convergence of minimizing paths to obtain confinement.

Suppose a final assertion in (ii) fails along the sequence. Take failing minimizers Q_k, parameterized on $[0, 1]$ at constant weighted speed L_k. The positive minimum of w bounds their original

speeds, so a subsequence converges uniformly to Q. Weighted-distance continuity gives $L_k \to L$, and Q is a weighted minimizer whose original endpoint distance is at least b.

The stronger construction for (D, E) supplies times $u < v$, a ratio $\widehat{c} < c'$, and a concrete shortcut S inside $A_{1,3}$ such that

$$(v-u)L \geq 2c\,, \qquad \operatorname{len}(S; w \cdot E) \leq \widehat{c}(v-u)L\,.$$

Recover S for E_k by Lemma 2.4, and attach short paths to the moving endpoints $Q_k(u), Q_k(v)$. The recovered weighted lengths have upper limit at most $\operatorname{len}(S; w \cdot E)$. For large k, the corresponding Q_k-segment has length at least c and the shortcut ratio is less than c', a contradiction.

The connection bound in (i) also persists, by the same recovery argument and its reserved strict margin. These arguments apply to every choice of endpoints and minimizing path. The collar estimate identifies the local and global threshold tests whenever their values are below the collar bound. □

5 Extreme ratios at many scales

Let (A_h, B_h) denote either ordered pair $(D_h, \widetilde{D}_h)$ or $(\widetilde{D}_h, D_h)$, and suppose

$$\mathsf{l} A_h \leq B_h \leq \mathsf{U} A_h\,, \qquad 0 < \mathsf{l} \leq \mathsf{U} < \infty\,.$$

We use the absolute scale lattice $\mathcal{S} = \{64^j : j \in \mathbb{Z}\}$. For each positive threshold below the optimal upper comparison constant, we will show that regular local witnesses have probability at least $p_* > 0$ at three quarters of the scales in each sufficiently small logarithmic window. The proof replaces annular crossings along a geodesic whenever too many scales fail this witness condition, thereby improving the global upper comparison.

5.1 Regular witnesses

Choose a small failure probability $\delta > 0$, to be fixed below. Scale tightness, internal-diameter tightness, and the local modulus give constants

$$0 < a < A_0 < \infty\,, \qquad 0 < \ell < a/100\,, \qquad 0 < \beta_0 < 1$$

and local events $F_r(z)$ with probability at least $1 - \delta$ such that, in units of $s_r(z)$:

(R1) the A_h-crossing distance of $A_{r,2r}(z)$ is at least a, and the internal A_h-diameter of $\overline{A}_{r,2r}(z)$ in $A_{3r/4,3r}(z)$ is at most A_0;

(R2) for each metric, the internal distance from $\overline{A}_{r,2r}(z)$ to either boundary of $A_{3r/4,3r}(z)$, measured in $A_{r/2,5r}(z)$, is greater than $4\max\{1, \mathsf{U}\}\ell$;

(R3) points of $\overline{A}_{r,2r}(z)$ at Euclidean distance less than $\beta_0 r$ have internal A_h-distance less than $\ell/4$ in $A_{3r/4,3r}(z)$.

Here $\overline{A}_{a,b}(z) = \{x : a \leq |x - z| \leq b\}$. First choose positive crossing and collar bounds and a finite diameter bound. Then choose ℓ to satisfy (R2), and finally choose β_0 by the upper modulus. These events are local modulo constants.

For $t \in (0, \mathsf{U})$, let $H_r(t)$ be the event $F_r(0)$ together with existence of a path $P \subset \overline{A}_{r,2r}$ which is an internal A_h-geodesic in $A_{3r/4,3r}$ and satisfies

$$\ell/2 \leq s_r(0)^{-1} \operatorname{len}(P; A_h) \leq \ell\,, \qquad B_h(P(0), P(1); A_{3r/4,3r}) > t \operatorname{len}(P; A_h)\,. \tag{5}$$

The closed-annulus convention is used throughout this section and in the fine-field trials below. It includes the boundary contacts of crossing paths. The collars identify both endpoint distances in (5) with their global versions, and (R3) gives endpoint separation at least $\beta_0 r$. Let $H_{z,r}(t)$ be the translated event; translation covariance gives the same probability.

Lemma 5.1 (Crossing improvement)**.** *On $F_r(z) \setminus H_{z,r}(t)$, every global A_h-geodesic subpath crossing $A_{r,2r}(z)$ while staying in its closure has endpoints at B_h-distance at most t times its A_h-length.*

Proof. Its length L lies in $[a, A_0]s_r(z)$. Divide it into $\lceil L/(\ell s_r(z))\rceil$ equal-length pieces. Every piece lies in the closed annulus, has length in $[\ell/2, \ell]s_r(z)$, and is an internal geodesic in the larger annulus. Absence of (5) gives the required bound on each piece. Add the bounds. □

Lemma 5.2 (Annular covering)**.** *Fix $\rho \in (0,1)$ and finite $L > 0$. Suppose a deterministic set $T \subset \mathcal{S} \cap [\varepsilon^2 R, \varepsilon R]$ has at least $\rho \log_{64} \varepsilon^{-1} - O(1)$ elements. Suppose $E_{z,r}$ is determined modulo constants by the field in $A_{r/2,5r}(z)$ and has probability at least p, uniformly over z, r. For p sufficiently close to one, with polynomially high probability every point of B_{LR} belongs to a ball $B_{r/2}(z)$ for which*

$$r \in T\,, \qquad z \in \frac{r}{100d}\mathbb{Z}^d\,, \qquad E_{z,r} \text{ occurs}\,.$$

Any prescribed polynomial power is obtained by taking p closer to one. This choice of p is independent of L and the additive constant in the lower bound for $|T|$; those quantities affect only the prefactor and the threshold for ε. All estimates are uniform in R.

Proof. For each point x of $(\varepsilon^2 R/(1000d))\mathbb{Z}^d \cap B_{(L+1)R}$, let $z_r(x)$ be a nearest point of $(r/(100d))\mathbb{Z}^d$. The event $E_{z_r(x),r}$ is measurable modulo constants from the field in $A_{r/4,8r}(x)$. In particular, the difference between the two centering spherical averages is measurable there.

For a desired power $s > 0$, choose $A > (2d+s)\log(64)/\rho$. The shell input (E4) gives a threshold p for which the probability of no successful scale at a fixed x is at most $e^{-A|T|}$. If $|T| \geq \rho \log_{64} \varepsilon^{-1} - C_0$, a union bound gives

$$C_L e^{AC_0} \varepsilon^{A\rho/\log 64 - 2d} = O_{L,C_0}(\varepsilon^s)\,.$$

The threshold p depends on A and the fixed annular geometry, but not on L, C_0. The rounding errors are much smaller than $r/2$, so the successful balls cover the points between grid sites. □

Choose δ and $p_* > 0$ small enough that

$$\mathbb{P}[F_r(z)] \geq 1 - \delta\,, \quad \mathbb{P}[H_r(t)] < p_* \quad \Longrightarrow \quad \mathbb{P}[F_r(z) \setminus H_{z,r}(t)] \geq p$$

for the threshold in Lemma 5.2 with $\rho = 1/4$. These choices and the regularity constants are independent of t. Independence of the threshold from L allows the macroscopic confinement radius to be chosen later.

Definition 5.3. For $u < \mathsf{U}$ and $\beta \in (0,1)$, let $G_R(u,\beta)$ be the event that some $x, y \in \overline{B}_{2R}$ satisfy

$$|x - y| \geq \beta R\,, \qquad \beta s_R(0) \leq A_h(x,y) \leq \beta^{-1} s_R(0)\,, \qquad B_h(x,y) > u A_h(x,y)\,.$$

Proposition 5.4 (Frequency of witnesses)**.** *There is $\theta > 0$, depending only on the metrics and regularity choices, such that if*

$$0 < t < \mathsf{U}\,, \qquad u > \mathsf{U} - \theta(\mathsf{U} - t)\,, \qquad \mathbb{P}[G_R(u,\beta)] \geq \beta\,, \tag{6}$$

then, for every sufficiently small $\varepsilon > 0$, at least three quarters of the scales in $\mathcal{S} \cap [\varepsilon^2 R, \varepsilon R]$ satisfy

$$\mathbb{P}[H_r(t)] \geq p_*\,. \tag{7}$$

For fixed t, u, β, the threshold for ε is uniform in R.

Proof. Set $\theta = a/A_0$. Suppose at least one quarter of the scales fail (7). Lemma 5.2 covers B_{LR} by balls for which $F_r(z) \setminus H_{z,r}(t)$ occurs, except on an event whose probability tends to zero uniformly in R. Choose L so that confinement of all relevant geodesics to B_{LR} fails with probability less than $\beta/10$.

Take a selected geodesic P with endpoints as in $G_R(u,\beta)$. Discard its prefix before its last visit to $\overline{B}_{4\varepsilon R}(x)$. On a common modulus event this prefix has length $o_\varepsilon(s_R(0))$, since a geodesic prefix has length equal to the distance between its endpoints. The same observation controls suffixes ending near y, irrespective of their Euclidean traces.

At a retained time t_i, choose a covering ball $B_{r_i/2}(z_i)$ containing $P(t_i)$, and let t_{i+1} be the next exit time from $B_{2r_i}(z_i)$. Both macroscopic endpoints are outside this ball. Write $I_i = [t_i, t_{i+1}]$, let $\widehat{t}_i$ be the last visit to $\partial B_{2r_i}(z_i)$ before t_i, and put $J_i = [\widehat{t}_i, t_{i+1}]$. By (R1) and the geodesic property,

$$\operatorname{len}(P|_{I_i}; A_h) \le \operatorname{len}(P|_{J_i}; A_h) \le A_0 s_{r_i}(z_i)\,.$$

Since $P(t_i) \in B_{r_i/2}(z_i)$, the last crossing of $\overline{A}_{r_i,2r_i}(z_i)$ before t_{i+1} lies in I_i. Its length is at least $a s_{r_i}(z_i)$ and, by Lemma 5.1, it admits a B_h replacement with ratio at most t.

Repeat from the exit point, stopping after the first step which passes the first subsequent visit to $\overline{B}_{4\varepsilon R}(y)$. The final exit lies within $8\varepsilon R$ of y, so the remaining suffix also has length $o_\varepsilon(s_R(0))$. There are only finitely many possible scales and centers, and every processed step has length bounded below by a positive random minimum. The procedure terminates after finitely many steps.

The forward intervals I_i are disjoint apart from their endpoints. Their replaced subintervals occupy at least a θ fraction of their total length; the auxiliary backward extensions J_i are used only for the length bound above. Use $B_h \le \mathsf{U} A_h$ on the other portions to obtain

$$B_h(x,y) \le [\mathsf{U} - \theta(\mathsf{U} - t)] A_h(x,y) + o_\varepsilon(s_R(0))\,. \tag{8}$$

The error is uniform over endpoints on the common modulus event. Since $A_h(x,y) \ge \beta s_R(0)$, this contradicts $G_R(u,\beta)$ for sufficiently small ε. The confinement, covering, and modulus failures together have probability less than β, contradicting (6). □

Corollary 5.5. *If* U *is the optimal essential upper comparison constant, then for every* $0 < t < \mathsf{U}$*, the scales satisfying* (7) *occupy at least three quarters of every sufficiently small window* $\mathcal{S} \cap [\varepsilon^2 R_0, \varepsilon R_0]$*, for some deterministic* $R_0 > 0$*. The same* R_0 *may be chosen for any finite list of applications.*

Proof. Choose u strictly between $\mathsf{U} - \theta(\mathsf{U} - t)$ and U. Optimality and continuity give a positive-probability ratio witness at deterministic rational endpoints. Enclose them in a deterministic ball. Positivity and finiteness of their normalized distance give some $\beta > 0$ with $\mathbb{P}[G_{R_0}(u,\beta)] \ge \beta$. Enlarging R_0 only changes the permissible β. Apply Proposition 5.4. □

6 A finite family of discounts

Assume $c_* < C_*$. Prepare the regularity constants of Section 5 for both ordered pairs, using the smaller of their two p_* values. Choose

$$c_* < c_0 < c_1 < c_2 < c_3 < C_*\,. \tag{9}$$

First choose c_1, c_2, c_3, and then choose c_0 close enough to c_* that

$$\frac{1}{c_0} > \frac{1}{c_*} - \theta_- \left(\frac{1}{c_*} - \frac{1}{c_1} \right), \tag{10}$$

where θ_- is the frequency constant for $(A_h, B_h) = (\widetilde{D}_h, D_h)$. Write $H_r^-(t)$ for its regular event. On $H_r^-(1/c_1)$, the witness is a favorable $\widetilde{D}_h$-geodesic arc satisfying

$$\operatorname{len}(\gamma; \widetilde{D}_h) < c_1 D_h(\gamma(0), \gamma(1)) \,,$$

with fixed normalized lower endpoint separation and upper length bounds.

Lemma 6.1 (Fine-field amplification)**.** *Fix finitely many disjoint deterministic open boxes in $A_{1,3}$ and smaller compact subboxes with nonempty interiors. For each $p < 1$, there is $\rho = 64^{-m}$ such that, whenever*

$$\mathbb{P}[H_{\rho r}^-(1/c_1)] \geq p_* \,, \tag{11}$$

with probability at least p every dilated compact subbox contains a simple rectifiable arc γ satisfying

$$\begin{gathered} \operatorname{len}(\gamma; \widetilde{D}_h) \leq c_2 D_h(\gamma(0), \gamma(1)) \,, \\ D_h(\gamma(0), \gamma(1)) \geq a_1 s_r(0) \,, \qquad \operatorname{len}(\gamma; D_h) \leq A_1 s_r(0) \,, \end{gathered} \tag{12}$$

for deterministic $0 < a_1 < A_1 < \infty$. All choices are uniform over r satisfying (11).

Proof. Step 1: Fine and coarse fields. Use the compact-kernel white-noise representation of [1, Section 2.3]. For a smooth radial compactly supported K with $\int K^2 = 1$, the LGF is represented modulo constants by

$$h(x) = \int_0^\infty \int_{\mathbb{R}^d} t^{-(d+1)/2} K((x-y)/t)\, W(dy, dt) \,.$$

At trial scale $s = \rho r$, split $h = F + g$, where $F = h_{<Ls}$ is the stationary fine field and g is smooth modulo constants. The fine field and the coarse field modulo constants are independent. If R_K bounds the support radius of K, the restriction $F|_{B_{5s}(w)}$ depends only on white noise with $t < Ls$ and spatial coordinate in $B_{(5+R_K L)s}(w)$. Disjoint such neighborhoods give independent fine-field restrictions.

For each fixed trial geometry, $\operatorname{osc}_{B_{5s}(w)} g \to 0$ in probability as $L \to \infty$, uniformly in s, w. Indeed, a derivative of order $k \geq 1$ has variance $O((Ls)^{-2k})$. After rescaling the trial ball and subtracting the value at its center, the expected squared Sobolev norm of sufficiently high fixed integer order is $O(L^{-2})$. Sobolev embedding gives the assertion. Choose $\delta > 0$ so that

$$e^{4\xi\delta} c_1 < c_2 \,, \qquad e^{4\xi\delta} < 2 \,.$$

Step 2: A local trial event. We use a trial event expressed in terms of curve lengths, since geodesicity need not survive a field perturbation. Write $\ell = \ell_-$ and set

$$I = \overline{A}_{s,2s}(w) \,, \quad O = A_{3s/4,3s}(w) \,, \quad J_0 = A_{s/2,5s}(w) \,, \quad C_{\mathrm{col}} = 4 \max\{1, c_*^{-1}\} \ell \,.$$

For an input X, put $s_X = s^{\xi Q} e^{\xi X_s(w)}$. Let $W_{w,s}(X)$ be the event that

$$D_X(I, \partial O; J_0) > e^{-\xi\delta} C_{\mathrm{col}} s_X \tag{13}$$

and there is a rectifiable curve $P \subset I$, with endpoints u, v, such that

$$\begin{gathered} D_X(u, v; O) \geq e^{-\xi\delta} \frac{\ell}{2c_1} s_X \,, \qquad \operatorname{len}(P; \widetilde{D}_X) \leq e^{\xi\delta} \ell s_X \,, \\ \operatorname{len}(P; \widetilde{D}_X) \leq e^{2\xi\delta} c_1 D_X(u, v; O) \,. \end{gathered} \tag{14}$$

These conditions are measurable from $X|_{J_0}$ and its spherical average there; curve lengths can be computed using the internal J_0 coordinate.

On $\{\operatorname{osc}_{J_0} g \le \delta\}$, the normalized fields $h - h_s(w)$ and $F - F_s(w)$ differ by a continuous function of absolute value at most δ on J_0. Weyl comparison therefore changes each normalized internal distance and curve length by a factor between $e^{-\xi\delta}$ and $e^{\xi\delta}$. A full-field $H^-_{w,s}(1/c_1)$ witness transfers to (13)–(14) for F. Hence

$$\mathbb{P}[W_{w,s}(F)] \ge p_* - \mathbb{P}[\operatorname{osc}_{J_0} g > \delta]\,.$$

The fine field is an LGF plus the possibly dependent smooth function $-g$, so the weak axioms apply. Locality makes this a function of the restricted fine field, unchanged by adding constants. The trial events are thus independent when their fine-noise neighborhoods are disjoint.

Step 3: Amplification in every box. Let M be the number of boxes. Choose N with $M(1 - p_*/2)^N < (1-p)/3$. Next choose L so large that each coarse-oscillation failure probability is less than both $p_*/2$ and $(1-p)/(3MN)$. Finally choose $\rho = 64^{-m}$ small enough to fit N trial regions, with disjoint fine-noise neighborhoods, inside each compact subbox. Independence of the fine trials and a separate union bound on coarse oscillations imply that every box has a successful witness and all coarse oscillations are small with probability at least $1 - 2(1-p)/3$.

Step 4: Return to the full field. Transfer a successful curve back to h. The resulting bounds are

$$\begin{aligned} D_h(u,v;O) &\ge e^{-2\xi\delta}\frac{\ell}{2c_1}s_h\,, \qquad \operatorname{len}(P;\widetilde{D}_h) \le e^{2\xi\delta}\ell s_h\,,\\ \operatorname{len}(P;\widetilde{D}_h) &\le e^{4\xi\delta}c_1 D_h(u,v;O)\,, \qquad D_h(I,\partial O;J_0) > e^{-2\xi\delta}C_{\mathrm{col}}s_h\,. \end{aligned} \tag{15}$$

The curve has D_h-length at most $c_*^{-1}e^{2\xi\delta}\ell s_h$, smaller than the collar bound. A competing path leaving O must pay this collar before its first exit. Thus $D_h(u,v) = D_h(u,v;O)$, and the strict inequality $e^{4\xi\delta}c_1 < c_2$ gives the favorable global ratio.

The finitely many Gaussian differences $h_s(w) - h_r(0)$ have variance $\log(1/\rho) + O(1)$ uniformly in r. Bound their absolute values by a deterministic K with failure probability less than $(1-p)/3$. Since $s_h/s_r(0) = \rho^{\xi Q}e^{\xi(h_s(w)-h_r(0))}$, we can use

$$a_1 = \rho^{\xi Q}e^{-\xi K - 2\xi\delta}\frac{\ell}{2c_1}\,, \qquad A_1 = \rho^{\xi Q}e^{\xi K + 2\xi\delta}\frac{\ell}{c_*}\,,$$

increasing A_1 if needed. Minimize $\widetilde{D}_h$-length inside the compact image of each witness. The minimizer exists and is simple, and it preserves the endpoint ratio. Its D_h-length is bounded by c_*^{-1} times its $\widetilde{D}_h$-length. This proves (12). □

Proposition 6.2 (Finite deterministic family)**.** *For each $p \in (0,1)$, there are $m \ge 1$, $\rho = 64^{-m}$, a thin deterministic open collector $V \Subset A_{1,3}$ containing ∂B_2, a finite family $f_1, \dots, f_J \in C_c^\infty(A_{1,3})$ which are nonnegative and positive exactly on V, and constants*

$$a, b, c, g, H > 0\,, \qquad c' < C_*\,, \qquad \max_j \|f_j\|_\infty \le M_f < \infty$$

with the following properties. Write $f_{z,r,j}(x) = f_j((x-z)/r)$, $V_{z,r} = z + rV$, and

$$\mathcal{R} = \{r \in \mathcal{S} : \mathbb{P}[H^-_{\rho r}(1/c_1)] \ge p_*\}\,. \tag{16}$$

For each $r \in \mathcal{R}$ and $z \in \mathbb{R}^d$, there are events $E_{z,r,j}$ determined modulo constants by the field in $A_{r/2,5r}(z)$ such that

$$\mathbb{P}\left[\bigcup_{j=1}^{J} E_{z,r,j}\right] \ge p\,.$$

On $E_{z,r,j}$, writing $s = s_r(z)$:

(T1) every path from $\overline{V}_{z,r}$ to either boundary of $A_{r,3r}(z)$ has original D_h-length at least as;

(T2) any two points of $\overline{V}_{z,r}$ have internal $D_{h-f_{z,r,j}}$-distance in $A_{r,3r}(z)$ at most $(2a-g)s$, where $g = a$;

(T3) consider an internal $D_{h-f_{z,r,j}}$-geodesic in $B_{4r}(z)$ meeting $\overline{V}_{z,r}$, and let u, v be its first and last collector points. If $D_h(u,v) \geq bs$, its portion between those visits contains a subsegment of length at least cs whose endpoints are joined in $B_{4r}(z)$ by a path of $\widetilde{D}_{h-f_{z,r,j}}$-length at most c' times that length;

(T4) if Z is a finite set of labeled sites containing (z,j) with pairwise disjoint radius- $4r$ balls, and $F = \sum_{(w,i)\in Z} f_{w,r,i}$, then every D_{h-F}-geodesic has at most one excursion into $B_{4r}(z)$ meeting $\overline{V}_{z,r}$. Its total original D_h-length spent in $V_{z,r}$ is at most Hs, with $H = ae^{\xi M_f}$.

The constants satisfy

$$2b < g/2\,. \tag{17}$$

For collector endpoints, the threshold in (T3) is unchanged if the original endpoint distance is replaced by the internal distance in $B_{4r}(z)$.

Proof. Step 1: Collars and access paths. We make the choices in normalized coordinates. Start with $V_0 = A_{7/4,9/4}$. Tightness and nondegeneracy give $a > 0$ such that, with probability as close to one as needed, the U-coordinate distances from $\overline{V}_0$ to $\partial A_{1,3}$ exceed $4a$. This choice precedes the graph and the depth of the discount. Set $g = a$, choose $0 < b < a/4$, and choose $\epsilon_0 < e^{-\xi}b/200$ with $3\epsilon_0 < a/4$.

Common upper moduli allow a sufficiently fine deterministic mesh of ∂B_2 and a thin collector $V \Subset V_0$ such that, except on an event of arbitrarily small probability, each enlarged vertex star has U-coordinate D-diameter at most $b/16$, and

$$\max_{x\in\overline{V}} \min_{v \text{ a mesh vertex}} D(x,v;U_1) \leq \epsilon_0/2\,. \tag{18}$$

Choose small central compact boxes inside the tree edge tubes, and slightly larger open boxes still inside those tubes for routing. Apply Lemma 6.1 to the compact boxes with sufficiently high success probability. This fixes ρ and gives common a_1, A_1 at every scale in (16). These constants need no further smallness or largeness restrictions.

Step 2: A compact set of good configurations. Take a compact family $\mathcal{C}$ of normalized internal coordinates on U, U_1 with sufficiently high probability. Within $\mathcal{C}$, impose the closed conditions: collars at least $4a$; star diameters at most $b/16$; (18); and in each central compact box, existence of a path $\gamma : [0,1] \to U$ with D^U-Lipschitz constant at most A_1, endpoint D^U-distance at least a_1, and

$$\operatorname{len}(\gamma; \widetilde{D}^U) \leq c_2 D^U(\gamma(0), \gamma(1))\,.$$

Call the resulting set $\mathcal{G}$. Allocating the preceding failure probabilities makes its probability at least p. It is compact. The path fibers are compact by the common inverse modulus, their length inequalities are closed by lower semicontinuity, and projection of these compact fibers preserves closedness. The remaining constraints are continuous inequalities on fixed compact sets.

Every configuration in $\mathcal{G}$, including configurations occurring only as limits, supplies the needed paths. By Lemma 2.2, the access inequalities give paths in $\overline{U}_1 \subset U$. A shortest $\widetilde{D}^U$-path in the compact image of a favorable witness is simple and preserves its endpoint ratio. Its D^U-length is

at most $c_*^{-1}c_2A_1$. Lemma 4.1, with this finite bound, builds the marked graph inside V. It uses the local length structure on U and leaves the ratio margin from c_2 to c_3.

The quantitative collar estimate preceding Lemma 4.2 localizes the construction. Any weighted path between collector points that exits $A_{1,3}$ pays at least $4a$ after its last collector visit, whereas access-and-graph competitors cost less than $3\epsilon_0 < a/4$. Take the graph weight small enough that $\lambda_n \operatorname{diam}(K, d_K) \le \epsilon_0/2$; this reserves a uniform strict margin for path recovery. All paths used in collapse and in shortcut transfer remain in a compact subset of $A_{1,3} \subset U$. Distance threshold tests below b agree for the global and inspected internal metrics by the same first-exit argument.

Step 3: Selection of finitely many discounts. Choose $c_3 < c' < C_*$. For each configuration in $\mathcal{G}$, Lemma 4.2 gives a smooth discount, connection costs below $a/2$, and a shortcut with endpoint threshold $b/2$, strict ratio below c', and a positive length margin. For this fixed discount the conclusions persist on a relative neighborhood in $\mathcal{C}$, retaining the strict connection bound below $a/2$, with final threshold b, a smaller positive length constant, and collar constant a. A finite subcover of $\mathcal{G}$ gives deterministic $f_1, \dots, f_J$. Take the minimum of their positive length constants for c and the maximum of their amplitudes for M_f. Membership in $\mathcal{C}$ and the corresponding neighborhoods defines the Borel events $E_{z,r,j}$. Their union contains $\mathcal{G}$.

To apply the local shortcut statement to an internal B_4-geodesic, restrict it between its first and last collector points. The connection competitor bounds this portion's weighted length by a number less than $a/2$. Exiting $A_{1,3}$ costs at least a, so this portion and every competitive local path stay in $A_{1,3}$. It is therefore a minimizing path for the inspected local weighted structure. The original endpoint threshold $b < a/4$ agrees with its global counterpart by the below-collar test. This proves (T1)–(T3).

Step 4: The occupation bound. To prove (T4), consider simultaneous discounts in disjoint radius- $4r$ balls. The segment of a discounted geodesic between its first and last visits to $\overline{V}_{z,r}$ has weighted length at most as, using the competitor from (T2). Other discounts do not affect this competitor. If two distinct $B_{4r}(z)$ excursions met the collector, the portion between them would pay one unaffected collar on departure and another on return. Restrict these portions between their collector visits and $\partial B_{3r}(z)$. All discounts vanish on their interiors, and (T1) makes their combined cost at least $2as$, a contradiction.

On the whole first-to-last collector segment, the weight inside $V_{z,r}$ is at least $e^{-\xi M_f}$. Lemma 2.1 therefore bounds the total original length there by $ae^{\xi M_f}s$. Only H depends on the discount depth; (17) was fixed earlier. □

Corollary 6.3 (Many usable scales)**.** *For all sufficiently small ε, uniformly over R satisfying*

$$\mathbb{P}[H_R^-(1/c_0)] \ge p_*\,, \tag{19}$$

at least two thirds of the scales in $\mathcal{S} \cap [\varepsilon^2 R, \varepsilon R]$ belong to $\mathcal{R}$.

Proof. The regular event in (19) gives $G_R(1/c_0, \beta)$ with probability at least β for a fixed $\beta > 0$. Condition (10) permits Proposition 5.4 with target $1/c_1$, giving three-quarters density of good arc scales. Passing from s to $r = s/\rho$ shifts the logarithmic index by the fixed integer m. At most $2m + O(1)$ indices are lost at the endpoints. For small ε, the remaining density is at least two thirds. □

7 Change of measure

Let $\mathcal{H}$ be the Cameron–Martin space of the LGF modulo constants. On smooth compactly supported functions,

$$\|f\|_{\mathcal{H}}^2 = a_d \int_{\mathbb{R}^d} |\zeta|^d |\widehat{f}(\zeta)|^2 \, d\zeta \,, \qquad a_d > 0 \,. \tag{20}$$

Translation and dilation preserve this critical norm. The fixed family in Proposition 6.2 therefore satisfies $\|f_{z,r,j}\|_{\mathcal{H}} \le M_0$ for some $M_0 < \infty$, uniformly in z, r, j. A sum F_Z of at most k bumps satisfies

$$\|F_Z\|_{\mathcal{H}}^2 \le M_0^2 k^2 \,. \tag{21}$$

No orthogonality is used. The quadratic dependence on k is sufficient because the count below fixes k before sending ε to zero.

Lemma 7.1 (Gaussian shift)**.** *For a measurable event A, a deterministic $F \in \mathcal{H}$, and $p_H > 1$,*

$$\mathbb{P}[A(h)] \le \exp\left\{ \frac{p_H - 1}{2} \|F\|_{\mathcal{H}}^2 \right\} \mathbb{P}[A(h+F)]^{1-1/p_H} \,. \tag{22}$$

Proof. Let μ be the Gaussian law modulo constants and μ_F the law of $h + F$. Its density is

$$L_F = \frac{d\mu_F}{d\mu} = \exp\{W(F) - \tfrac{1}{2}\|F\|_{\mathcal{H}}^2\} \,,$$

where $W(F)$ is centered Gaussian with variance $\|F\|_{\mathcal{H}}^2$. Hölder's inequality under μ_F gives

$$\mu(A) \le \mu_F(A)^{1-1/p_H} \left(\mathbb{E}_{\mu_F} L_F^{-p_H} \right)^{1/p_H} .$$

The expectation is $\exp\{p_H(p_H - 1)\|F\|_{\mathcal{H}}^2/2\}$. The equivalent measures have the same completed null sets, so the inequality also holds in the completed sigma-algebra. □

Lemma 7.2 (A common covering event)**.** *Choose p in Proposition 6.2 sufficiently close to one. For every R satisfying (19), with probability tending to one polynomially as $\varepsilon \to 0$, every point of B_{5R} belongs to a ball $B_{r/2}(z)$ with*

$$r \in \mathcal{R} \cap [\varepsilon^2 R, \varepsilon R] \,, \qquad z \in \frac{r}{100d} \mathbb{Z}^d \,, \qquad E_{z,r,j} \text{ for some } j \,. \tag{23}$$

The rate is uniform in R. This event is chosen before the endpoints and the number of perturbation sites.

Proof. Corollary 6.3 gives a fixed positive density of usable scales. Apply Lemma 5.2 to the union of local events at each such scale. The probability threshold is independent of the fixed logarithmic shift m, which affects only how small ε must be. □

8 Counting perturbations

We compare two counts of finite sets of discounts. A geodesic whose endpoint ratio is close to C_* gives many sets for which the discounted geodesic makes substantial collector passages. After changing the field, all such sets are tested against one common base geodesic, and the shortcut and occupation estimates bound their number. This follows the comparison architecture of [4,

Section 4]. We give the full count because both bounds depend on the graph construction and the finite choice of discounts.

Fix $0 < \eta \le 1$ and deterministic $x, y \in B_{3R}$ with $|x - y| \ge \eta R$. Throughout this section, $0 < \varepsilon < 1/3$ and all perturbation scales satisfy $r \in \mathcal{R} \cap [\varepsilon^2 R, \varepsilon R]$. Put

$$S = R^{\xi Q} e^{\xi h_{8R}(0)}\,, \qquad D = D_h(x, y)\,, \qquad \widetilde{D} = \widetilde{D}_h(x, y)\,. \tag{24}$$

Each bump used below is supported in B_{6R}, so it leaves S unchanged. Bumps at a common scale with centers separated by at least $10r$ also leave each other's spherical normalizers $s_r(z)$ unchanged.

Lemma 8.1 (Bounds for local normalizers)**.** *There is a deterministic $B > 0$ such that, uniformly in R, with probability tending to one polynomially as $\varepsilon \to 0$,*

$$\varepsilon^B \le s_r(z)/S \le \varepsilon^{-B} \tag{25}$$

for all $r \in \mathcal{S} \cap [\varepsilon^2 R, \varepsilon R]$ and $z \in (r/(100d))\mathbb{Z}^d \cap B_{5R}$.

Proof. The Gaussian variable $h_r(z) - h_{8R}(0)$ is centered and has variance at most

$$\log(R/r) + O(1) \le 2 \log \varepsilon^{-1} + O(1)\,.$$

This follows by integrating the logarithmic covariance against the two spherical measures. The smaller sphere stays a fixed positive multiple of R away from ∂B_{8R}. There are $O(\varepsilon^{-2d} \log \varepsilon^{-1})$ variables. Gaussian tails and a union bound bound their absolute values by $B_0 \log \varepsilon^{-1}$ with polynomially high probability for sufficiently large B_0. Take $B > 2\xi Q + \xi B_0 + 1$. □

At each scale, partition $(r/(100d))\mathbb{Z}^d$ into $(1000d)^d$ colors, each with spacing $10r$. The radius- $4r$ balls centered at points of one color are disjoint. Fix a scale, a color, and a dyadic number $q \in [\varepsilon^B/2, \varepsilon^{-B}]$. A labeled set has the form

$$Z = \{(z_1, j_1), \dots, (z_n, j_n)\}\,, \qquad F_Z = \sum_{i=1}^{n} f_{z_i, r, j_i}\,,$$

where the centers are distinct points of that color in B_{5R} and $x, y \notin \overline{B}_{4r}(z_i)$. Let $\mathcal{F}_Z(h)$ be the event:

(C1) $\widetilde{D} \ge C_* D - qS$;

(C2) each E_{z_i, r, j_i} occurs and $s_r(z_i) \in [qS, 2qS]$;

(C3) the selected geodesic P_h visits $B_r(z_i)$ for every i;

(C4) for every i, the selected $P_{h - F_Z}$ has a $B_{4r}(z_i)$ excursion meeting $\overline{V}_{z_i, r}$ whose first and last collector points have original internal D_h-distance in $B_{4r}(z_i)$ at least $b s_r(z_i)$.

Lemma 8.2 (Separated shifts)**.** *For a fixed labeled set Z, put $f_i = f_{z_i, r, j_i}$. Almost surely, for every i,*

$$(h + F_Z)_r(z_i) = h_r(z_i)\,, \qquad (h + F_Z)_{8R}(0) = h_{8R}(0)\,,$$

and, for either metric and every open $W \subset B_{5r}(z_i)$,

$$D_{h + F_Z}(\cdot, \cdot; W) = D_{h + f_i}(\cdot, \cdot; W)\,.$$

Consequently,

$$E_{z_i,r,j_i}(h + F_Z) = E_{z_i,r,j_i}(h + f_i)\,, \qquad P_{(h+F_Z)-F_Z} = P_h\,.$$

Conditions (C1)–(C4) are invariant under adding a constant to the field. The same conclusions hold when shifted fields are returned to the representative with unit-sphere average zero.

Proof. Each f_i vanishes on its own radius- r sphere. A bump at another center is zero throughout $B_{5r}(z_i)$ by the separation of centers and its support in a radius- $3r$ ball. Also, all supports lie in B_{6R} and miss ∂B_{8R}. This proves the average identities.

On W, the function $F_Z - f_i$ vanishes. Weyl scaling and Lemma 2.1 therefore identify the internal metrics. Their normalized annular coordinates agree, so the local event identities follow. Subtracting F_Z from $h + F_Z$ gives the selected path identity.

Adding a constant a multiplies all distances, lengths, and normalizers by $e^{\xi a}$ and leaves the selected geodesic unchanged. Thus each counting condition is invariant. In particular, discounting $h + F_Z - (F_Z)_1(0)$ gives $h - (F_Z)_1(0)$, whose selected geodesic is still P_h. □

The internal identity also holds for negative shifts. If a $B_{4r}(z_i)$ excursion of P_{h-F_Z} meets the collector, the segment between its first and last closed collector visits is compactly contained in $B_{4r}(z_i)$. It minimizes length internally there, and its internal metric is that for $h - f_i$. Hence (T3) applies without treating a path with endpoints on the boundary of an open ball as an internal geodesic. With base field $h + F_Z$, this gives the same local use of the fixed path P_h in the shifted count.

The bumps and their finite sums belong to a countable deterministic family. Their almost-sure identities hold on one common event. The measurable selection and compact path descriptions in Section 2.3 make all counting events universally measurable, which suffices for Lemma 7.1.

Lemma 8.3 (Pruning)**.** *If a set of n labeled sites satisfies (C2)–(C3), it has a subset Z' of at least κn sites satisfying (C4), for a deterministic $\kappa > 0$. If (C1) also holds, then $\mathcal{F}_{Z'}(h)$ occurs.*

Proof. Write $s_i = s_r(z_i)$ and

$$\Delta(Z) = D_h(x, y) - D_{h-F_Z}(x, y)\,.$$

For each site, take a component of $P_h \cap B_{2r}(z_i)$ containing a visit to $B_r(z_i)$. Its endpoints lie on $\partial B_{2r}(z_i)$, and its original length is at least $2as_i$ by the two collars. Replace it by the discounted connection in (T2). These intervals are disjoint, and elsewhere the discounts only shorten the path. Thus

$$\Delta(Z) \geq g \sum_{i \in Z} s_i \geq gnqS\,. \tag{26}$$

If a site fails (C4) for the current discounted geodesic, remove that discount. By (T4), at most one excursion into its radius- $4r$ ball meets the collector. In that excursion, replace the first-to-last collector segment by an original internal path of cost less than bs_i, as permitted by failure of (C4). It avoids every other bump. If no excursion meets the collector, the discount contributes no path cost. Removing the site therefore increases the distance by at most $bs_i \leq 2bqS$.

Continue until all surviving sites satisfy (C4). By (17), the final set satisfies

$$\Delta(Z') \geq (g - 2b)nqS \geq (g/2)nqS\,.$$

Restoring the original weight along its discounted geodesic and using (T4) gives

$$\Delta(Z') \leq H \sum_{i \in Z'} s_i \leq 2H|Z'|qS\,.$$

Hence $|Z'| \geq \kappa n$ with $\kappa = g/(4H) = e^{-\xi M_f}/4$. The original local events and visits of P_h are unchanged by pruning. □

Lemma 8.4 (Many unshifted sets)**.** *On the common covering event of Lemma 7.2 and the normalizer event of Lemma 8.1, if*

$$\widetilde{D} \geq C_* D - \varepsilon^{B+1} S\,, \tag{27}$$

then, for all sufficiently small ε, simultaneously for every integer $\lceil 1/\kappa \rceil \leq k \leq \lfloor \varepsilon^{-1/2} \rfloor$,

$$\sum_{r,\ \text{color},\ q} \sum_{1 \leq |Z| \leq k} \mathbf{1}_{\mathcal{F}_Z(h)} \geq (\varepsilon^{-1/2}/k)^{\lceil \kappa k \rceil}\,. \tag{28}$$

The outer sum has $O(\log^2 \varepsilon^{-1})$ terms. The threshold for ε is uniform in R, x, y subject to the fixed relative separation.

Proof. Take the portion of P_h first crossing radii $\eta R/4$ to $\eta R/2$ about x. At first hits of levels separated by $20\varepsilon R$, choose covering balls from (23). Their centers are separated by more than $8\varepsilon R$, lie in B_{5R}, and stay a positive multiple of R away from both endpoints for small ε. There are at least $c_\eta \varepsilon^{-1}$ centers. Each is visited by P_h in its radius- r ball and has a valid label.

Pigeonholing the scale, color, and dyadic bin for $s_r(z)/S$ leaves at least $N \geq \varepsilon^{-1/2}$ sites with common r, q. Their radius- $4r$ balls are disjoint. Since $q \geq \varepsilon^B/2$, (27) implies (C1). This choice of ε is independent of k.

Every k-subset of these sites prunes to a favorable subset of size at least $m = \lceil \kappa k \rceil$ and at most k. A fixed output is contained in at most $\binom{N-m}{k-m}$ input sets. The number of distinct outputs is at least

$$\frac{\binom{N}{k}}{\binom{N-m}{k-m}} = \frac{(N)_m}{(k)_m} \geq (N/k)^m\,,$$

where $(a)_m = a(a-1)\cdots(a-m+1)$. This gives (28). □

Lemma 8.5 (Few shifted sets)**.** *For fixed r, q and color, outside a common null set,*

$$\sum_{1 \leq |Z| \leq k} \mathbf{1}_{\mathcal{F}_Z(h+F_Z)} \leq K_0^k \tag{29}$$

for a deterministic $K_0 < \infty$ independent of r, q, R, x, y, k.

Proof. Call (z, j) very good if $E_{z,r,j}(h + f_{z,r,j})$ occurs, $s_r(z) \in [qS, 2qS]$, and P_h has an excursion satisfying (C4) when its original internal distance is measured by $D_{h+f_{z,r,j}}$. If $\mathcal{F}_Z(h + F_Z)$ occurs, every label in Z is very good by Lemma 8.2.

If no shifted event occurs, there is nothing to prove. Otherwise fix one such Z_0 with $|Z_0| \leq k$. At each very good center outside Z_0, apply (T3) to P_h, viewed locally as the discounted geodesic for the field $h + f_{z,r,j}$. Choose one type per center. This gives a subsegment of D_h-length at least cqS and a $\widetilde{D}_h$-shortcut of ratio at most c'. The subsegments are disjoint in time. Their balls miss all supports of F_{Z_0}, so the shortcuts remain valid for $h + F_{Z_0}$.

Apply the occupation bound with base field $h + f_{z,r,j}$ at the sites of Z_0. It gives

$$\operatorname{len}(P_h; D_{h+F_{Z_0}}) \leq D_h(x, y) + H \sum_{i \in Z_0} s_i \leq D_{h+F_{Z_0}}(x, y) + 2Hkq S\,.$$

If there are N very good centers outside Z_0, use their shortcuts and the bound $\widetilde{D}_{h+F_{Z_0}} \le C_* D_{h+F_{Z_0}}$ on the remaining portions. The near-extreme condition for $\mathcal{F}_{Z_0}(h+F_{Z_0})$ yields

$$(C_* - c')cqS\,N \le 2C_* Hkq S + qS\,.$$

Thus the total number of very good centers is at most $K_1 k$. Each has at most J labels, and each shifted favorable set is a subset of those labels. There are at most $2^{K_1 Jk}$ such subsets. □

Proposition 8.6 (A truncated near-extreme bound)**.** *Suppose R satisfies (19), and let $\mathcal{A}_{\varepsilon R}$ be the intersection of the common covering and normalizer events. For every $N > 0$,*

$$\mathbb{P}\left[\mathcal{A}_{\varepsilon R},\ \widetilde{D}_h(x,y) \ge C_* D_h(x,y) - \varepsilon^{B+1} S\right] = O_N(\varepsilon^N)\,. \tag{30}$$

The estimate is uniform over these R and deterministic endpoint pairs with the prescribed positive relative separation. The event $\mathcal{A}_{\varepsilon R}$ is independent of the endpoints and of N.

Proof. For a fixed scale, color, and bin, there are at most $C_d\varepsilon^{-2d}$ possible centers. The number M of labeled sets of size at most k is at most

$$M \le (1 + JC_d\varepsilon^{-2d})^k \le (C\varepsilon^{-2d})^k\,.$$

For instance, encode each set by an ordered list of length k padded with a dummy symbol.

Apply Lemma 7.1 separately to each event $\mathcal{F}_Z$ and its own shift F_Z, then apply Hölder's inequality to the finite sum. Equations (21) and (29) give

$$\begin{aligned}\sum_Z \mathbb{P}[\mathcal{F}_Z(h)] &\le e^{A_p k^2} M^{1/p_H} \left(\sum_Z \mathbb{P}[\mathcal{F}_Z(h+F_Z)]\right)^{1-1/p_H} \\ &\le \exp\{A_p k^2 + C_p k\}\varepsilon^{-2dk/p_H}\,, \qquad A_p = \tfrac{1}{2}(p_H - 1)M_0^2\,.\end{aligned}$$

The fields in the shifted indicators share the same base h; the shifts themselves may differ.

Sum over the $O(\log^2 \varepsilon^{-1})$ outer choices and use Lemma 8.4. For $\lceil 1/\kappa \rceil \le k \le \lfloor \varepsilon^{-1/2} \rfloor$,

$$\mathbb{P}[\mathcal{A}_{\varepsilon R},\ (27)] \le C_0 \log^2 \varepsilon^{-1}\ \exp\{A_p k^2 + C_p k\}\varepsilon^{-2dk/p_H}(k\sqrt{\varepsilon})^{\lceil \kappa k \rceil}\,. \tag{31}$$

The threshold for ε in the counting lemmas is independent of k.

Choose $p_H > 8d/\kappa$ after fixing the discount family, and set $b_0 = \kappa/2 - 2d/p_H > \kappa/4$. Given N, fix an integer $k \ge \lceil 1/\kappa \rceil$ with $b_0 k > N + 1$. For sufficiently small ε, this fixed k is admissible. The right side of (31) is bounded by $C_{k,p_H} \log^2 \varepsilon^{-1} \varepsilon^{b_0 k} = O_N(\varepsilon^N)$. The quadratic Cameron–Martin cost is a finite constant because k was fixed first. □

9 Closing the comparison interval

Apply Section 5 to $(D_h, \widetilde{D}_h)$ and write its regular event as $H_R^+(t)$. Its constants ℓ_+, β_+ do not depend on $t < C_*$. We will find a deterministic $t < C_*$ such that

$$\mathbb{P}[H_R^+(t)] < p_* \quad \text{for every } R \text{ satisfying (19)}\,. \tag{32}$$

This contradicts the frequency conclusion for the two ordered pairs on a common sufficiently small window.

Choose $L < \infty$ such that, uniformly in R,

$$\mathbb{P}[L^{-1} \le s_R(0)/S \le L] > 1 - p_*/10\,. \tag{33}$$

The logarithm of this ratio is a centered Gaussian with fixed variance. Choose $0 < \alpha < \xi(Q - \sqrt{2d})$ and ν with $\nu\alpha > B + 2$. On the common upper-modulus event, rounding an endpoint in B_{2R} to an $\varepsilon^\nu R$ grid changes its D_h-distance by at most $LO(\varepsilon^{\nu\alpha})S$ on (33); the change for $\widetilde{D}_h$ is at most C_* times as large. This modulus event has probability tending to one uniformly in R.

For a sufficiently small ε to be fixed below, set

$$t = C_* - \frac{\varepsilon^{B+1}}{4\ell_+ L}\,. \tag{34}$$

On $H_R^+(t)$ the witness endpoints satisfy $|x - y| \ge \beta_+ R$ and $D_h(x, y) \le \ell_+ s_R(0)$. Thus on (33) their additive comparison gap is at most $\varepsilon^{B+1}S/4$. Their nearest grid points x', y' lie in B_{3R} and are separated by at least $\beta_+ R/2$ for small ε. The triangle inequality gives

$$\begin{aligned} C_* D_h(x', y') - \widetilde{D}_h(x', y') &\le C_* D_h(x, y) - \widetilde{D}_h(x, y) \\ &\quad + 2C_*\big(D_h(x, x') + D_h(y, y')\big) \le \varepsilon^{B+1} S\,. \end{aligned}$$

The final inequality holds because $\nu\alpha > B + 2$.

There are $O(\varepsilon^{-2\nu d})$ grid pairs. Choose $N > 2\nu d + 1$ in Proposition 8.6, including its fixed choice of k, and then take ε sufficiently small. The probability that any such pair satisfies the near-extreme inequality on $\mathcal{A}_{\varepsilon R}$ is $O_N(\varepsilon^{N-2\nu d})$. The covering, bin, modulus, and macroscopic-normalizer failures are common to all endpoint pairs, so their probabilities are added once. Their total, together with the grid-pair bound, is less than p_* for a fixed sufficiently small ε, uniformly over R satisfying (19). This proves (32).

Keep this ε, and hence t, fixed. Apply Corollary 5.5 to the lower ordered pair with threshold $1/c_0$ and to the upper ordered pair with threshold t. Choose a common deterministic macroscopic radius R_0. For a new parameter $\delta \downarrow 0$, let

$$\mathcal{S}_\delta = \mathcal{S} \cap [\delta^2 R_0, \delta R_0]\,.$$

For all sufficiently small δ, each of the lower-good and upper-good sets occupies at least three quarters of $\mathcal{S}_\delta$. This finite set is nonempty, so the two sets intersect. At any scale in their intersection, (32) contradicts upper goodness.

We have ruled out $c_* < C_*$. Hence $\widetilde{D}_h = c_* D_h$ almost surely for all endpoint pairs. Simultaneous Weyl scaling extends this equality, with the same deterministic constant, to every continuous perturbation. This is the proposed proof of Theorem 1.4.

10 Covariance and convergence

This section gives the consequences of Theorem 1.4. These deductions are conditional on the preceding uniqueness argument.

Lemma 10.1 (Strong metrics are weak)**.** *Every assignment satisfying Definition 1.1 belongs to the weak class, with scale constants $r^{\xi Q}$.*

Proof. For an LGF, exact covariance and Weyl scaling give

$$r^{-\xi Q} e^{-\xi h_r(0)} D_h(r\cdot, r\cdot) = D_{h(r\cdot)-h_r(0)} .$$

The field on the right has the law of the LGF normalized by its unit-sphere average. These metric laws, and the laws of internal metrics on fixed domains, are therefore identical for all $r > 0$.

Now replace h by $h+f$, with f a possibly dependent random continuous function. In normalized coordinates the Weyl weight is $e^{\xi(f(r\cdot)-f_r(0))}$. For $0 < r < R$ and on a fixed compact set, this weight and its reciprocal have a finite random common bound, obtained from the supremum of $|f|$ on a larger compact set. Local upper bounds and moduli follow from the identical-law unweighted family. For a lower modulus, a path from its initial point to a Euclidean-separated endpoint must first exit a small ball. Up to that exit the weight is bounded below, and the unweighted crossing cost has a uniform positive lower modulus. Subsequent travel outside the compact set cannot remove this cost.

These upper and lower moduli on a compact exhaustion give the required tightness and nondegeneracy. If f is bounded, the global factor $e^{2\xi\|f\|_\infty}$ gives the same bounds for all $r > 0$. □

Corollary 10.2 (Exact dilation covariance)**.** *Assume Theorem 1.4. Every weak assignment satisfies, for each fixed $r > 0$, almost surely for all x, y,*

$$D_h(rx, ry) = D_{h(r\cdot)+Q\log r}(x, y) .$$

The identity holds after every continuous perturbation of h.

Proof. For fixed $r > 0$, define

$$(T_r D)_h(x, y) = r^{-\xi Q} D_{h(\cdot/r)}(rx, ry) .$$

This satisfies the weak axioms. Locality, translation covariance, and Weyl scaling follow by a change of variables. If the scale constants of D are $\mathfrak{c}_s$, use $r^{-\xi Q}\mathfrak{c}_{rs}$ for $T_r D$; the normalized scale- s metric is the normalized scale- rs metric of D evaluated at $h(\cdot/r)$.

Weak uniqueness gives a deterministic $a(r) > 0$ with $T_r D = a(r)D$, including after continuous perturbations. The identity $T_s T_r = T_{sr}$ gives $a(rs) = a(r)a(s)$. If $h_1(0) = 0$, then

$$r^{-\xi Q} e^{-\xi h_r(0)} D_h(r\cdot, r\cdot) \quad \text{has the law of} \quad a(r) D_h .$$

Input (E2) forces deterministic bounds $0 < a_- \le a(r) \le a_+ < \infty$ for all $r > 0$. An unbounded sequence would violate tightness of a positive point distance; a sequence tending to zero would have the zero pseudometric as its limit. Apply these bounds to $a(r^n) = a(r)^n$ to obtain $a(r) = 1$. Evaluating $T_r D = D$ at the field $h(r\cdot)$ gives the claimed identity. Simultaneous Weyl scaling extends it to all continuous perturbations. □

Input (E1) supplies a weak assignment. Corollary 10.2 upgrades it to exact dilation covariance, and translation covariance is already a weak axiom. Lemma 10.1 makes any two strong assignments weak assignments. This proves the axiomatic part of Theorem 1.2, conditional on Theorem 1.4.

Lemma 10.3 (Support of a point distance)**.** *For an LGF normalized by $h_1(0) = 0$, the law of $D_h(0, e_1)$ gives positive probability to every nonempty open subinterval of $(0, \infty)$. In particular, this law has a unique median.*

Proof. Choose smooth compactly supported $0 \le f \le 1$, equal to one on a thin connected open neighborhood U of $[0, e_1]$, such that $f_1(0) < 1$. Put $g = f - f_1(0)$ and $a = 1 - f_1(0) > 0$. For each fixed real t, the normalized laws of $h + tg$ and h are equivalent by the Cameron–Martin theorem applied to tf modulo constants.

For every realization on the simultaneous Weyl event, the map $t \mapsto D_{h+tg}(0, e_1)$ is continuous, since g is bounded. Rectifiable connectivity of U gives

$$D_{h+tg}(0, e_1) \le e^{\xi a t} D_h(0, e_1; U) \longrightarrow 0 \quad \text{as } t \to -\infty\,.$$

Choose $B_\delta(0) \Subset U$ excluding e_1. Every path to e_1 pays at least

$$e^{\xi a t} D_h(0, \partial B_\delta(0)) \longrightarrow \infty \quad \text{as } t \to \infty$$

before its first exit. The unweighted boundary distance is strictly positive.

Every prescribed interval $(u, v) \Subset (0, \infty)$ is therefore hit for some rational t. The union of these shifted events over rational t has probability one, so one shifted event has positive probability. Equivalence of laws gives positive probability for the unshifted event. If $m < n$ were both medians, the inequalities $\mathbb{P}[D_h(0, e_1) \le m] \ge 1/2$ and $\mathbb{P}[D_h(0, e_1) \ge n] \ge 1/2$ would force $\mathbb{P}[m < D_h(0, e_1) < n] = 0$, contradicting full support. □

Corollary 10.4 (Full LFPP convergence)**.** *Assume Theorem 1.4. Fix a convolution kernel* K *from [1, Remark 1.1], and set* $\mathsf{K}_\epsilon(x) = \epsilon^{-d}\mathsf{K}(x/\epsilon)$ *and* $h^*_\epsilon = h * \mathsf{K}_\epsilon$*. Define*

$$D^\epsilon_h(x, y) = \inf_{P : x \to y} \int_0^1 e^{\xi h^*_\epsilon(P(t))} |P'(t)|\, dt\,,$$

where P is piecewise continuously differentiable. Let a_ϵ be any deterministic choice of median of $D^\epsilon_h(0, e_1)$ under $h_1(0) = 0$. Then $\mathsf{a}_\epsilon^{-1} D^\epsilon_h$ converges in probability locally uniformly as $\epsilon \to 0$ to the unique weak, hence strong, assignment for which $D_h(0, e_1)$ has median one. With the same normalizers, convergence holds for each LGF plus a bounded continuous function.

Proof. By input (E1), every sequence tending to zero has a deterministic subsequence converging in probability to a weak assignment. The closed-half-line inequalities of the portmanteau theorem show that one is a median of each limiting point distance. Weak uniqueness makes any two such limits deterministic multiples. Lemma 10.3 gives a unique median, so the multiple is one.

Fix one subsequential limit. If full convergence in probability failed, there would be a sequence staying a positive probability away from this limit. It would have a further subsequence converging in probability to the same limit, a contradiction. Input (E1) supplies one subsequence assignment for each LGF plus a bounded continuous function; simultaneous Weyl scaling identifies all these assignments with the same deterministic scalar. □

A Gaussian shell decoupling

We give a direct proof of the high-probability shell estimate used in Lemma 5.2. This proof uses the covariance of the LGF and Gaussian Hölder's inequality; it does not require a radial Markov property.

Lemma A.1 (Shell decoupling)**.** *Fix $0 < a < 1 < b$, $q \in (0, 1)$, and radii $r_{i+1} \le q r_i$. Let E_i be measurable with respect to $(h - h_{r_i}(0))|_{A_{ar_i, br_i}}$. For each $A > 0$ and $\theta \in (0, 1)$, there exists $p < 1$, depending only on these parameters and d, such that*

$$\mathbb{P}[E_i] \ge p \text{ for every } i \quad \Longrightarrow \quad \mathbb{P}\left[\sum_{i=1}^n \mathbf{1}_{E_i} < \theta n\right] \le e^{-An} \quad \text{for all } n \ge 1\,.$$

Proof. For a compactly supported zero-mass distribution ν with finite logarithmic energy, write

$$\|\nu\|_-^2 = \operatorname{Var}(h,\nu) = \kappa_d \int_{\mathbb{R}^d} |\zeta|^{-d} |\widehat{\nu}(\zeta)|^2 \, d\zeta \,.$$

This norm is invariant under mass-preserving dilation. A linear observable of the normalized field on the ith annulus has the form (h,ν_i), where

$$\nu_i = \varphi_i - \left(\int \varphi_i \right) \sigma_{r_i} \,, \qquad \varphi_i \in C_c^\infty(A_{ar_i,br_i}) \,.$$

Here σ_r is uniform probability measure on ∂B_r. These measures have finite energy, and the corresponding pairings with sphere measures are obtained by Gaussian L^2 approximation. After rescaling, the supports lie in a fixed compact annulus away from zero.

For two such unit-scale zero-mass distributions ν, μ and sufficiently small ϵ, we claim

$$|\operatorname{Cov}((h,\nu_\epsilon),(h,\mu))| \le C\epsilon \|\nu\|_- \|\mu\|_- \,. \tag{35}$$

The covariance kernel is $\log|v-\epsilon u|^{-1}$. Since ν has zero mass, subtract $\log|v|^{-1}$. Multiply by smooth compact cutoffs equal to one on the two supports, with the v cutoff away from zero. The resulting kernel $G_\epsilon(u,v)$ has C^M norm at most $C_M\epsilon$ for every fixed integer M, because $|v-\epsilon u|$ is bounded below.

Periodize on a fixed sufficiently large cube. The Fourier coefficients satisfy

$$|a_{m,n}| \le C_M \epsilon (1+|m|+|n|)^{-M} \,.$$

For a further cutoff χ equal to one on the supports, Fourier duality and Cauchy–Schwarz give

$$|\nu(\chi e^{im\cdot u})| \le C\|\nu\|_- \|\chi e^{im\cdot u}\|_{\dot H^{d/2}} \le C'\|\nu\|_- (1+|m|)^{d/2} \,.$$

Use the analogous bound for μ and take $M > 3d$. The double Fourier series is absolutely summable after these polynomial factors, proving (35). Approximation extends the estimate to the closed Gaussian Hilbert spaces generated by the annular fields.

For nearby logarithmic scales use Cauchy–Schwarz. Only finitely many index differences fail the separation needed for (35); for the others the bound decays as $Cq^{|i-j|}$. Consequently every finite Gaussian vector assembled from these blocks has covariance matrix T satisfying

$$T \le K \operatorname{diag}(T_{11},\dots,T_{nn})$$

for a $K \ge 1$ independent of the number of blocks and coordinates. To see this, apply the scalar covariance bounds to arbitrary linear combinations in each block and bound the resulting quadratic form by the maximum absolute row sum.

Gaussian Hölder's inequality [7, Theorem 1(i)] gives, for nonnegative functions of finite block vectors,

$$\mathbb{E} \prod_i F_i(X_i) \le \prod_i \left(\mathbb{E}[F_i(X_i)^K]\right)^{1/K} \,.$$

For functions of the entire restricted fields, apply this to conditional expectations on increasing finite coordinate sets and use bounded martingale convergence. In particular, if $\mathbb{P}[E_i^c] \le \delta$, then

$$\mathbb{P}\left[\bigcap_{i\in I} E_i^c\right] \le \delta^{|I|/K}$$

for each index set I. A union bound over possible failing indices yields

$$\mathbb{P}\left[\sum_{i=1}^{n} \mathbf{1}_{E_i} < \theta n\right] \leq 2^n \delta^{(1-\theta)n/K}.$$

Choose $\delta \leq \exp\{-K(A + \log 2)/(1-\theta)\}$ and $p = 1 - \delta$. □

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